

Project S-19-05

Research Notebook

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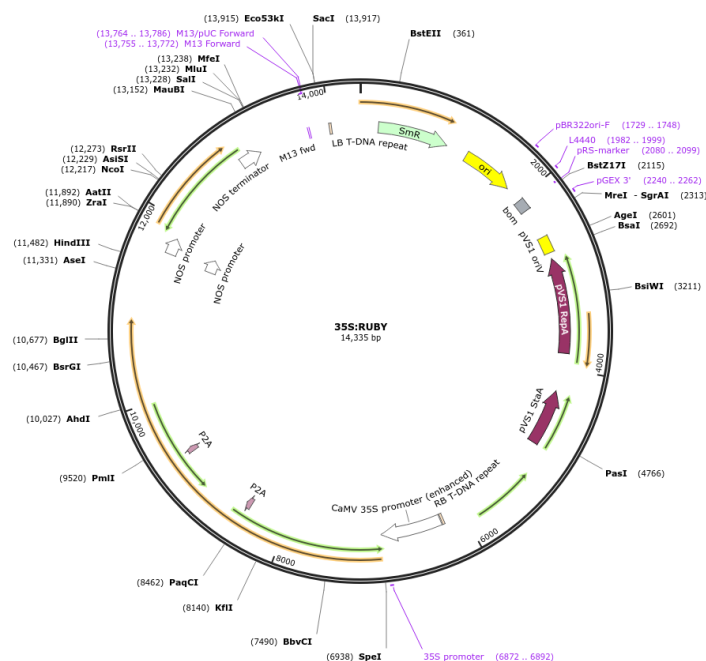
California Science & Engineering Fair

Agar Plate Preparation for Agrobacterium Growth and Bacterial Culture

January 27, 2025: Following project approval and ordering of materials, experimentation began. First, agar plate preparation for *Agrobacterium tumefaciens* growth and bacterial culture were performed. A 50mL conical tube was used to measure out 150mL of diH₂O into a glass bottle, where Luria Broth (LB) + Rifampin + Spectinomycin Agar (25mg/L (3.04μM) Rifampicin, 50mg/L (150.μM) Spectinomycin) (35g/L) was added into the diH₂O. The LB/Rifampin/Spectinomycin Agar diH₂O mixture was shaken vigorously. LB/Rifampin/Spectinomycin Agar was dissolved by microwaving for a few seconds at a time until agar was translucent, followed by cooling for 20 minutes. The plates were poured and cooled @ 20 C for 1 hour, then stored @ 4 C flipped.

- One day following, only one out of seven of the agar plates solidified, indicating inadequate agar concentration. The semi-solidified agar was reheated and repoured and subsequently solidified completely.

February 10, 2025: Lyophilized pre-transfected *A. tumefaciens* (CYP76AD1, DODA, UGT73A36, RUBY (SmR, pVS1 RepA, pVS1 StaA, P2A)) (ODIN) (RUBY Plasmid Map shown below) was hydrated with ~200µl(+) of diH2O using Pasteur pipette. The hydrated Agrobacterium was mixed until dissolved, where ~50µl of Agrobacterium was streaked onto 1 LB/Rifampin/Spectinomycin Agar Plate using an inoculation loop. Agrobacteria were grown for ~16-18 hours (overnight) in temperatures < 30 C.



February 13, 2025: After culturing transfected and hydrated *A. tumefaciens* on Luria Broth (LB) + Rifampin + Spectinomycin Agar (25mg/L (3.04μM) Rifampicin, 50mg/L (150μM) Spectinomycin) (35g/L) plates for six days @ 24 C, small, isolated brownish round entire flat punctiform colonies appeared after day four, suggesting prolonged bacterial lag time. Plant infection with this bacterial group was unsuccessful or qualitatively unapparent—suggesting low transformation efficiency. However, under the same conditions when grown on BD Difco YPD (yeast extract 10g/L, peptone 20g/L, dextrose 20g/L) broth, *A. tumefaciens* growth was characterized by large colonies (>5 mm) in half the time period on the selectively isolating nonfungal growth medium. YPD-*A. tumefaciens* was used for plant transformation on 2/14/25 to test for effective transformation.

Transformation / Plant Infection

February 10, 2025: Injection media (IM) (containing MES (2-(N-morpholino)ethanesulfonic acid) buffer, MgCl₂, and vanillin (acetosyringone (a plant phenolic compound that can induce the *vir* genes on the bacterial Ti plasmid) substitute)) (ODIN) was diluted by filling a 50mL tube with diH₂O where 1mL of concentrate IM was pipetted into the 50mL diH₂O tube (~2% IM). 4mL of the diluted IM (dIM) was added into a new 15mL tube. Using an inoculation loop, bacteria from the LB/Rifampin/Spectinomycin Agar Plate were gently scraped off and submerged into the 4mL tube of dIM (shown in illustration below). Bacteria were swirled and pipetted up and down to dissolve. Using a 1mL blunt syringe, 500μl was drawn from the 4mL Agrobacterium–dIM solution and placed firmly against a flat underside spot of the leaves of *Lactuca sativa* var. *capitata* BBP w/ Tyr, *Lactuca sativa* var. *capitata* BBP w/o Tyr, and *Nicotiana tabacum* BBP w/o Tyr, avoiding veins. The plunger was slowly depressed, saturating the leaf tissues with Agrobacterium–dIM solution, resulting in infection.

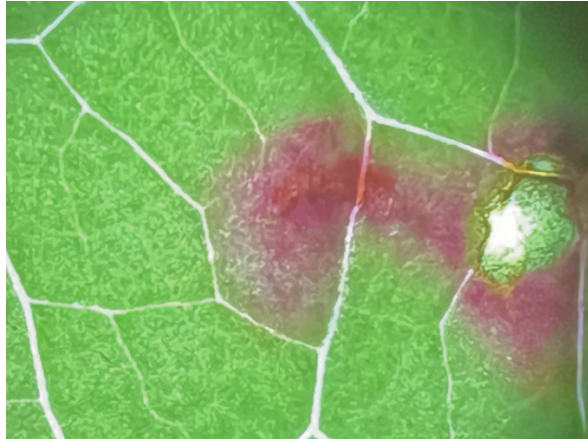
February 14-18, 2025: Temporal Qualitative Measures for BBP Reconstitution

~ Infection of *NtBBP* the bacteria grown on YPD media was performed on 2/14/25

- No visual sign of betanin production was seen on 2/17/25
- Minimal visual sign of betanin production was seen on 2/18/25

~ Infection of *LsvcBBPTyr* the bacteria grown on YPD media was performed on 2/15/25

- Visual sign of betanin production was seen on 2/16/25, with brighter reddish spots around the injection site and dissipating redness around the area of solution saturation
- Betanin product increased marginally up till 2/18/25



~ Infection of *LsvcBBP* the bacteria grown on YPD media was performed on 2/15/25

- Visual signs (much less that of *LsvcBBPTyr*) of betanin production were seen on 2/17/25, with reddish spots around the injection site and dissipating redness around the area of solution saturation (majority of production)
- Consistent on 2/18/25

The pinkish appearance plausibly caused by the oxidation of polyphenols of *LsvcBBPTyr* was ruled out with compound light microscopy and supported by the change of pH of the leaf tissue and qualitative observation that the hue of red most closely matches that of betanin. Assuming that the betalain biosynthesis pathway was indeed successful in all plants (just not to a visual degree), pH studies was conducted.

Non-Spectroscopic Acid-Base Techniques for the Quantification of the Hydrolytic Sensitivity of (2S)-1-{2-[(2S)-2,6-dicarboxy-2,3-dihydropyridin-4(1H)-ylidene]ethylidene}-5-(β-d-glucopyranosyloxy)-6-hydroxy-2,3-dihydro-1H-indol-1-ium-2-carboxylate in Wild-type and Transgenic Plant Models and Plant Transformation Efficiency

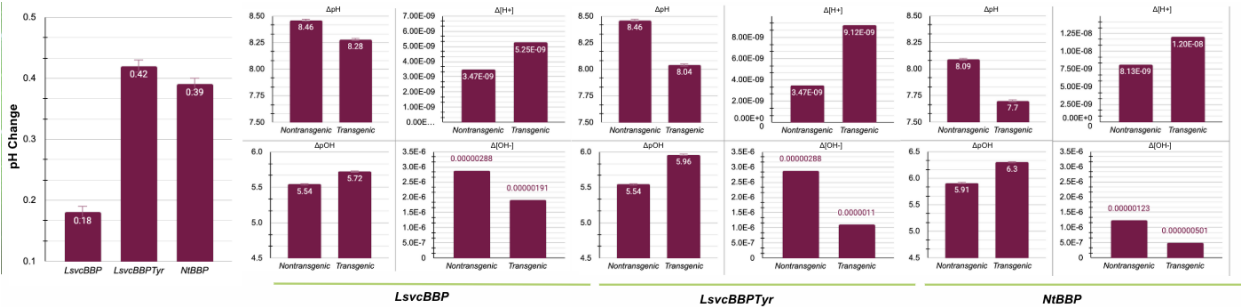
February 10/20, 2025: pH/pOH/[H⁺]/[OH⁻] were measured (with VIVOSON pH digital meter) and/or calculated with the lysed (mortar and pestle) plant leaf tissues samples (nontransgenic (2/10), transgenic (2/20)) + 21mL of H₂O (acts as a common neutralization buffer).

Acid-Base Parameters	Nontransgenic <i>LsvcBBP</i> Lysate + 21ml H ₂ O	Transgenic <i>LsvcBBP</i> Lysate + 21ml H ₂ O
pH	8.46 ± 0.01	8.28 ± 0.01
pOH	5.54	5.72 ± 0.01
[H ⁺]	3.47e-9	5.25e-9
[OH ⁻]	0.00000288	0.00000191

Acid-Base Parameters	Nontransgenic <i>LsvcBBPTyr</i> Lysate + 21ml H ₂ O	Transgenic <i>LsvcBBPTyr</i> Lysate + 21ml H ₂ O
pH	8.46 ± 0.01	8.04 ± 0.01
pOH	5.54	5.96 ± 0.01
[H ⁺]	3.47e-9	9.12e-9
[OH ⁻]	0.00000288	0.00000110

Acid-Base Parameters	Nontransgenic <i>NtBBP</i> Lysate + 21ml H ₂ O	Transgenic <i>NtBBP</i> Lysate + 21ml H ₂ O
pH	8.09 ± 0.01	7.70 ± 0.01
pOH	5.91	6.30 ± 0.01
[H ⁺]	8.13e-9	1.20e-8
[OH ⁻]	0.00000123	5.01e-7

Graphing the following data we get:



February 24, 2025: Calculations of hydrolytic sensitivity and plant infection efficiency:

$$\Delta\psi_{L.sativa} = \text{Transgenic pH} - \text{Nontransgenic pH}$$

$$\Delta\psi_{N.tabacum} = \text{Transgenic pH} - \text{Nontransgenic pH}$$

$$\Delta\Delta\psi = \Delta\psi_{N.tabacum} - \Delta\psi_{L.sativa}$$

$$W_{N.tabacum} = 82.5\%$$

$$W_{L.sativa} = 95.7\%$$

$$W_{N.tabacum} - W_{L.sativa} = 82.5 - 95.7 = -13.2$$

$$S = \frac{-\Delta\Delta\psi}{W_{N.tabacum} - W_{L.sativa}}$$

Where $\Delta\psi$ is the change in pH and W is water content. S (Hydrolytic Sensitivity) indicates that for every 1% increase in water content, there is a decrease of approximately $|S|$ pH units. (pH units per percent). The hydrolytic sensitivity in terms of pH units per percent decrease of C24H26N2O13 is S . The greater the $\Delta\psi$, the more betanin catalyzed from tyrosine. This novel quantification method leverages betanin's acidity, respective to the basic leaf tissue, quantifying pigment production without spectrophotometric techniques. W of plants was determined by FooDB chemical constituents list.

Regarding plant infection efficiency, *LsvcBBPTyr* had the greatest plant infection efficiency relating injected bacterial volume to betanin production in terms of pH per μL of agrobacteria: $B2B_{LsvcBBP} = 1.32 \pm 1.08 \text{ pH}_{\text{unit}} * \mu\text{LBV}^{-1}$, $B2B_{LsvcBBPTyr} = 3.08 \pm 2.52 \text{ pH}_{\text{unit}} * \mu\text{LBV}^{-1}$, $B2B_{LsvcBBP} = 2.86 \pm 2.34 \text{ pH}_{\text{unit}} * \mu\text{LBV}^{-1}$.

Maintaining Plant Health and Supplementation

Jan. 10 to Feb. 18, 2025: All *Lactuca sativa* var. *capitata* (Bonnie Plants) were grown and kept at 21 C at 40% humidity and watered with 400 mL of water every 1.5 weeks. *LsvcBBPTyr* was supplemented with 100 mL of 1M L-Tyrosine 4 days pre- and post-infection. *Nicotiana tabacum* (ODIN) plant was grown and kept at 21 C at 40% humidity and watered with 100 mL water once per week.

- Complications with Plant Growth (1/20): *LsvcBBPTyr* experienced a visual loss of turgor compared to *LsvcBBP* control, potentially caused by high-dose tyrosine absorption (eventually ruled out). Tyrosine supplementation developed a viscid white layer at the soil's humus of *LsvcBBPTyr*. Both *Lsvc* plants ended up declining and later dying for uncertain reasons-new plants were ordered and plant care and supplement/case regimen was replicated.